LAB ASSIGNMENT – 2

Name: DishaBiswas

Roll No: 204 (CSE)

DATE:19/01/2024

Q1. Make a schema first and then insert 6 documents.

- Roll_no=[1,2,3,4,5,]
- Name=["Ram","Alex","John","Bob","Mukesh","Danny"]
- Age=[20,19,40,55,30,28]
- Salary=[546.7,333.4,666.7,678.4,245.6,546.3]
- City=["A","B","C","D","E","F"]
- Phone_no=[123,456,122,444,567,892]

```
Query: db.students.insertMany([
... {roll_no: 1, name: "Ram",   age: 20, salary: 546.7, city: "A", phone_no: 123},
... {roll_no: 2, name: "Alex",  age: 19, salary: 333.4, city: "B", phone_no: 456},
... {roll_no: 3, name: "John",  age: 40, salary: 666.7, city: "C", phone_no: 122},
... {roll_no: 4, name: "Bob",   age: 55, salary: 678.4, city: "D", phone_no: 444},
... {roll_no: 5, name: "Mukesh", age: 30, salary: 245.6, city: "E", phone_no: 567},
... {roll_no: 6, name: "Danny", age: 28, salary: 546.3, city: "F", phone_no: 892}
... ])
Output: {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('697611febf3c95f1a01e2621'),
    '1': ObjectId('697611febf3c95f1a01e2622'),
    '2': ObjectId('697611febf3c95f1a01e2623'),
    '3': ObjectId('697611febf3c95f1a01e2624'),
    '4': ObjectId('697611febf3c95f1a01e2625'),
    '5': ObjectId('697611febf3c95f1a01e2626')
  }
}
```

Q2. Write a query to update the name of RAM to SAM. Query:

```
db.students.updateOne(
...  { name: "Ram" },
...  { $set: { name: "Sam" } }
... )
Output: {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q3. Write a query to display only the cities present in that collection. Query:

```
db.students.find({}, { city: 1, _id: 0 })
Output:[{ city: 'A' },
  { city: 'B' },
  { city: 'C' },
  { city: 'D' },
  { city: 'E' },
  { city: 'F' }]
```

Q4. Write a query to update the salary by 10%.

Query: db.students.updateMany({}, { $mul: { salary: 1.1 } }) Output: {

```
  acknowledged: true,
  insertedId: null,
  matchedCount: 6,
  modifiedCount: 6,
  upsertedCount: 0
}
```

Q5. Write a query to display all the documents in ascending and descending order of age. Query:

db.students.find().sort({age: 1})

Output: [{

_id: ObjectId('697611febf3c95f1a01e2622'),

roll_no: 2,

name: 'Alex', age:

19,

salary: 366.74,

city: 'B',

phone_no: 456

},

{

_id: ObjectId('697611febf3c95f1a01e2621'),

roll_no: 1,

name: 'Sam',

age: 20,

salary: 601.3700000000001,

city: 'A',

phone_no: 123

},

{

_id: ObjectId('697611febf3c95f1a01e2626'),

roll_no: 6,

name: 'Danny',

age: 28,

salary: 600.93,

city: 'F',

phone_no: 892

},

{

_id: ObjectId('697611febf3c95f1a01e2625'),

```
    roll_no: 5, name:
    'Mukesh', age: 30,
    salary: 270.16,
    city: 'E',
    phone_no: 567
  },
  {
    _id: ObjectId('697611febf3c95f1a01e2623'),
    roll_no: 3,
    name: 'John',
    age: 40,
    salary: 733.3700000000001,
    city: 'C',
    phone_no: 122
  },
  {
    _id: ObjectId('697611febf3c95f1a01e2624'),
    roll_no: 4,
    name: 'Bob', age:
    55,
    salary: 746.24,
    city: 'D',
    phone_no: 444
  }]
```

Query: db.students.find().sort({ age: -1 }) Output:

```
[
  {
    _id: ObjectId('697611febf3c95f1a01e2624'),
    roll_no: 4,
```

name: 'Bob', age:
55,
salary: 746.24,
city: 'D',
phone_no: 444
},
{
_id: ObjectId('697611febf3c95f1a01e2623'),
roll_no: 3,
name: 'John',
age: 40,
salary: 733.3700000000001,
city: 'C',
phone_no: 122
},
{
_id: ObjectId('697611febf3c95f1a01e2625'),
roll_no: 5,
name: 'Mukesh', age:
30,
salary: 270.16,
city: 'E',
phone_no: 567
},
{
_id: ObjectId('697611febf3c95f1a01e2626'),
roll_no: 6,
name: 'Danny',
age: 28,
salary: 600.93,
city: 'F',

```
  phone_no: 892
 },
 {
  _id: ObjectId('697611febf3c95f1a01e2621'),
  roll_no: 1,
  name: 'Sam',
  age: 20,
  salary: 601.3700000000001,
  city: 'A',
  phone_no: 123
 },
 {
  _id: ObjectId('697611febf3c95f1a01e2622'),
  roll_no: 2,
  name: 'Alex', age:
  19,
  salary: 366.74,
  city: 'B',
  phone_no: 456
 }]
```

Q6. Write a query to display all the documents with City → A,B,C. Query:

db.students.find({city: { $in: ["A", "B", "C"] }})

Output: [{

```
  _id: ObjectId('697611febf3c95f1a01e2621'),
  roll_no: 1,
  name: 'Sam',
  age: 20,
  salary: 601.3700000000001,
  city: 'A',
  phone_no: 123
 },
```

```
  {
    _id: ObjectId('697611febf3c95f1a01e2622'),
    roll_no: 2,
    name: 'Alex', age:
    19,
    salary: 366.74,
    city: 'B',
    phone_no: 456
  },
  {
    _id: ObjectId('697611febf3c95f1a01e2623'),
    roll_no: 3,
    name: 'John',
    age: 40,
    salary: 733.3700000000001,
    city: 'C',
    phone_no: 122
  }
]
```

Q7. Write a query to display only two documents from the entire collection. Query:

db.students.find().limit(2)

Output: [{

```
    _id: ObjectId('697611febf3c95f1a01e2621'),
    roll_no: 1,
    name: 'Sam',
    age: 20,
    salary: 601.3700000000001,
    city: 'A',
    phone_no: 123
  },
  {
```

_id: ObjectId('697611febf3c95f1a01e2622'),

roll_no: 2,

name: 'Alex', age:

19,

salary: 366.74,

city: 'B',

phone_no: 456

}]

Q8. Write a query to delete a document with ROLL_NO:5. Query:

db.students.deleteOne({ roll_no: 5 })

Output: { acknowledged: true, deletedCount: 1 }

Q9. Write a query to display all the documents with AGE greater than 20. Query:

db.students.find({ age: { $gt: 20 } })

Output: [{

_id: ObjectId('697611febf3c95f1a01e2623'),

roll_no: 3,

name: 'John',

age: 40,

salary: 733.3700000000001,

city: 'C',

phone_no: 122

},

{

_id: ObjectId('697611febf3c95f1a01e2624'),

roll_no: 4,

name: 'Bob', age:

55,

salary: 746.24,

city: 'D',

phone_no: 444

},

```
 {
   _id: ObjectId('697611febf3c95f1a01e2626'),
   roll_no: 6,
   name: 'Danny',
   age: 28,
   salary: 600.93,
   city: 'F',
   phone_no: 892
 }]
```

Q10. Write a query to display all the documents with AGE less than 20. Query:

db.students.find({age: {$lt: 20}})

Output:

```
[{
   _id: ObjectId('697611febf3c95f1a01e2622'),
   roll_no: 2,
   name: 'Alex', age:
   19,
   salary: 366.74,
   city: 'B',
   phone_no: 456
 }]
```

Q11. Write a query to display all the documents with AGE equals to 20. Query:

db.students.find({age: 20})

Output:

```
[{
   _id: ObjectId('697611febf3c95f1a01e2621'),
   roll_no: 1,
   name: 'Sam',
   age: 20,
   salary: 601.3700000000001,
   city: 'A',
   phone_no: 123
 }]
```

Q12. Write a query to display all the documents with AGE not equals to 20. Query:

db.students.find({age: {$ne: 20}})

Output: [

{

_id: ObjectId('697611febf3c95f1a01e2622'),

roll_no: 2,

name: 'Alex', age:

19,

salary: 366.74,

city: 'B',

phone_no: 456

},

{

_id: ObjectId('697611febf3c95f1a01e2623'),

roll_no: 3,

name: 'John',

age: 40,

salary: 733.3700000000001,

city: 'C',

phone_no: 122

},

{

_id: ObjectId('697611febf3c95f1a01e2624'),

roll_no: 4,

name: 'Bob', age:

55,

salary: 746.24,

city: 'D',

phone_no: 444

},

{

```
  _id: ObjectId('697611febf3c95f1a01e2626'),
  roll_no: 6,
  name: 'Danny',
  age: 28,
  salary: 600.93,
  city: 'F',
  phone_no: 892
 }
]
```

Q13. Write a query to display all the documents where AGE is greater than equals to 30. Query:

db.students.find({age: {$gte: 30}})

Output: [

```
 {
  _id: ObjectId('697611febf3c95f1a01e2623'),
  roll_no: 3,
  name: 'John',
  age: 40,
  salary: 733.3700000000001,
  city: 'C',
  phone_no: 122
 },
 {
  _id: ObjectId('697611febf3c95f1a01e2624'),
  roll_no: 4,
  name: 'Bob', age:
  55,
  salary: 746.24,
  city: 'D',
  phone_no: 444
 }
]
```

```
_id: ObjectId('697611febf3c95f1a01e2621')
roll_no : 1
name : "Sam"
age : 20
salary : 601.3700000000001
city : "A"
phone_no : 123
```

```
_id: ObjectId('697611febf3c95f1a01e2622')
roll_no : 2
name : "Alex"
age : 19
salary : 366.74
city : "B"
phone_no : 456
```

```
_id: ObjectId('697611febf3c95f1a01e2623')
roll_no : 3
name : "John"
age : 40
salary : 733.3700000000001
city : "C"
phone_no : 122
```

```
_id: ObjectId('697611febf3c95f1a01e2624')
roll_no : 4
name : "Bob"
age : 55
salary : 746.24
city : "D"
phone_no : 444
```

```
_id: ObjectId('697611febf3c95f1a01e2626')
roll_no : 6
name : "Danny"
age : 28
salary : 600.93
city : "F"
phone_no : 892
```